BLACK QUARTER

ABHIJITH SP
CVAS POOKODE

SUBSCRIBE

INTRODUCTIO N

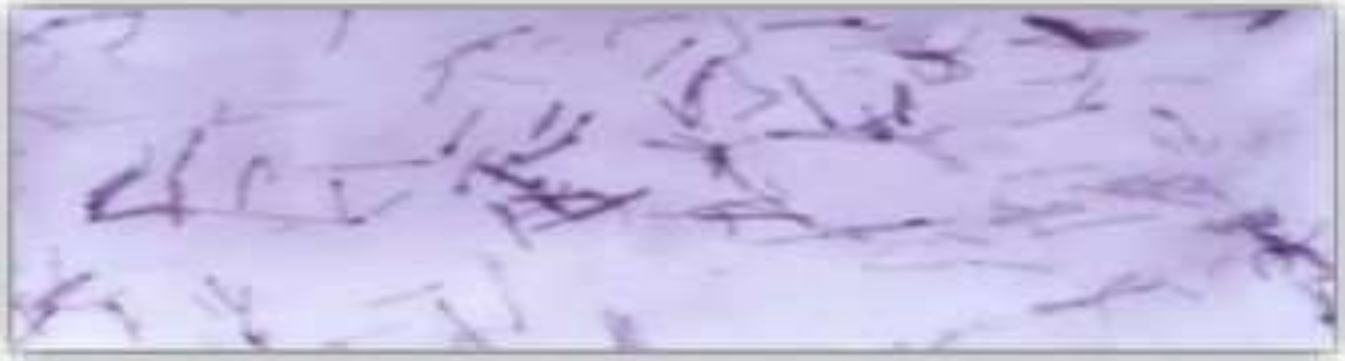
- An acute disease of cattle characterized by emphysematous swelling usually in heavy muscles.
- Buffaloes usually suffer from a milder form.
- Contaminated pasture appears to be major source of infection.
- Healthy animals in the age group 6 months to 2 years are generally affected.

BLACK QUARTER

- Blackleg is an acute, highly fatal, bacterial disease, mainly affecting cattle and buffaloes, between ages of 6 months to 2 years
- The disease is caused by Clostridium chauvoei,
- The characterized signs are the development of emphysematous swelling (puffy) in
- muscles.
- The name ‘blackleg’ derives from the fact that the infection mainly seen at the leg muscles, which become dark (black) in color, seen at PM.

AETIOLOGY

- Blackleg is most commonly caused by ***Clostridium chauvoei*****,** but ***C. feseri*** and other clostridial species can be isolated from some lesions.
- C. chauvoei is Gram-positive, rod-shaped, anaerobic, and motile, spore forming bacteria.
- The organisms probably are **ingested**, pass through the wall of the GI tract, and after gaining access to the **bloodstream**, are **deposited in muscle** and other tissues (spleen, liver, and alimentary tract) and may remain dormant indefinitely.



Etiology

- The disease organisms (*Cl. Chauvoei*) are spore forming, gas producing bacteria which can live
- in soil for years at dormant state. The disease is sporadic, mainly seen in cattle and buffaloes,
- Rarely in sheep and goats. More cases of disease are seen during or after the rain or flood season, particularly at semi-hilly areas.

SYNONYMS

- BLACK LEG
- QUARTER EVIL
- QUARTER ILL
- SYMPTOMATIC ANTHRAX

ANTIGENS & TOXINS

- *C. chauvoei* has somatic, spore (*C. chauvoei* shares a common spore antigen with *C. septicu*m) and flagellar antigens and produces 4 toxins. Toxins are highly pathogenic to mice & guinea pigs while poultry and rabbits are resistant to toxins.
- 1. Alpha toxin - oxygen stable haemolysin, necrotoxin that causes dermo necrosis and fibrinolysis
- 2. Beta toxin – DNAase
- 3. Gamma toxin – hyaluronidase
- 4. Delta toxin (chauveolysin) – Oxygen labile haemolysin.
- 5. Neuraminidase

Susceptible Host

- Common disease for cattle and sheep but occasionally goats and pigs also suffer from the disease.

TRANSMISSIO N

- Contaminated soil.
- The organisms enter into the digestive tract with feed and through cuts which occur during the shearing, docking, and castration, and via naval infection during birth.
- Infection of the vulva and vagina of the ewes during lambing may cause serious outbreak of the disease. Black leg is worldwide in distribution.
- Well nourished and grass fed animals are more often affected.

TRANSMISSION

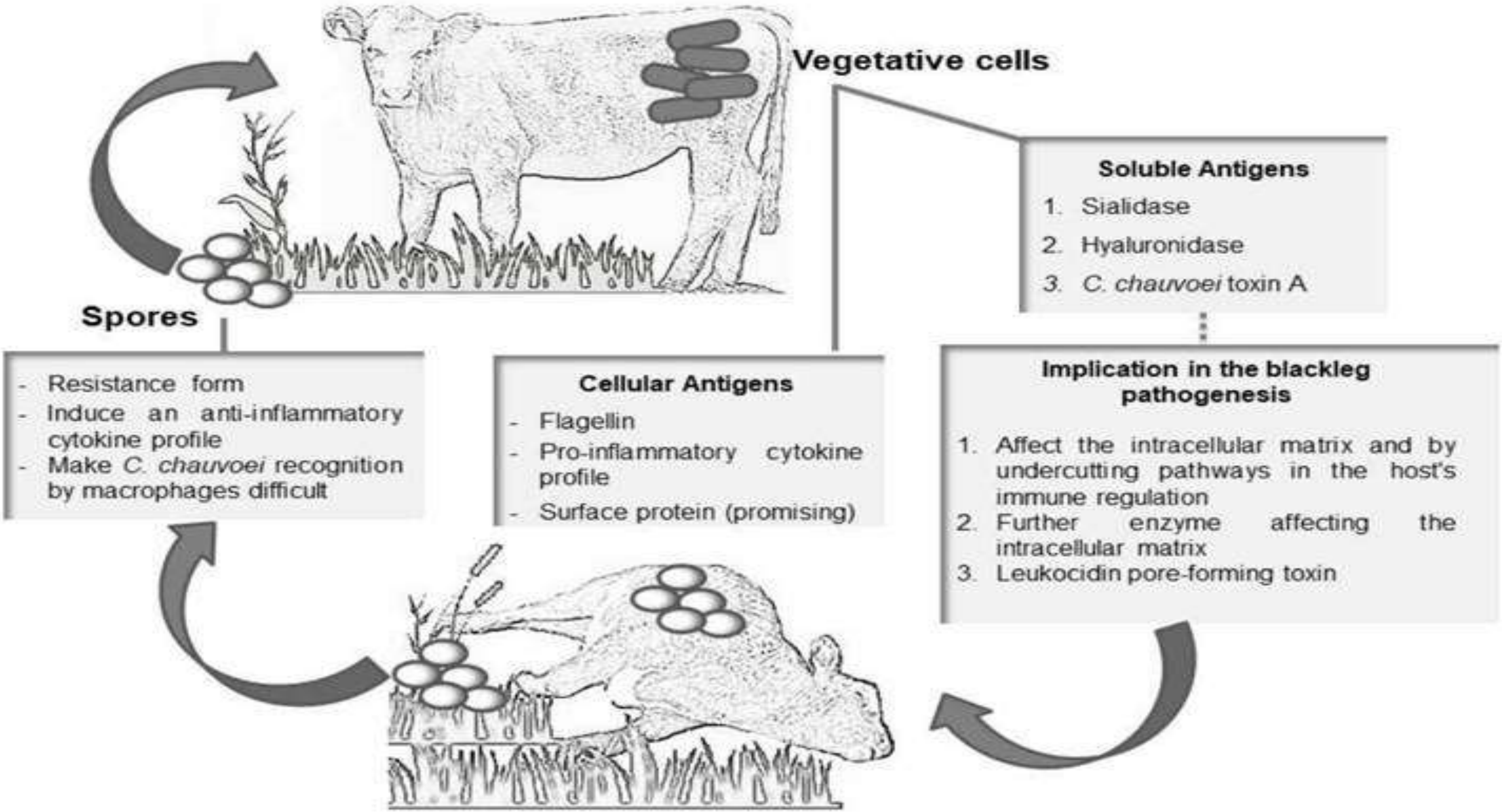
- Soil born
- Spores 1st enter the gastro-intestinal tract, along with contaminated soil/pasture
- Then blood and finally, in muscles of the body.
- Not transmitted from sick to healthy animals

INCUBATION PERIOD

- The **incubation period** extends from 12 months to several years. The animal aged 3 to 6 years mostly suffer from the disease.
- Affected animals may not show clinical symptoms continue to discharge organisms in faeces.
- The organisms persist in pastures for about 1 year.

PATHOGENESIS

- Ingestion from soil/Or through wounds.
- Multiply in the intestine and produce Bacteremia
- Enter skeletal muscle and remain dormant until anaerobic condition is produced.
- Then proliferate at an alarming rate and produce toxins that cause necrosis of muscle cells.
- Toxins are highly pathogenic to lab animals while poultry and rabbits are resistant.
- Alpha Toxin cause dermo necrosis and fibrinolysis.



Vegetative cells
Soluble Antigens
1. Sialidase
2. Hyaluronidase
3. C. chauvoei toxin A
Spores
- Resistance form
- Induce an anti-inflammatory cytokine profile
- Make C. chauvoei recognition by macrophages difficult
Cellular Antigens
- Flagellin
- Pro-inflammatory cytokine profile
- Surface protein (promising)
Implication in the blackleg pathogenesis
1. Affect the intracellular matrix and by undercutting pathways in the host's immune regulation
2. Further enzyme affecting the intracellular matrix
3. Leukocidin pore-forming toxin

CLINICAL SIGNS

- The disease usually occurs in young well fed cattle of 6 months to about 2-3 years of age.
- There are two forms of disease in cattle as acute and per acute form.
- In per acute there will be sudden death without symptoms. In acute fever is exhibited. The most obvious sign is crepitating swelling particularly in the hind or fore quarter which rackels when rubbed with the fingers as a result of gas production.
- Area will be cold, painless and black in colour. The affected animal will become lame and the affected muscles shows trembling with violent twitching. Death usually occurs within 24 hours.
- In sheep an acute febrile condition develops within 1-2 days following an injury and a typical black quarter lesion can be observed at the site. Death occurs suddenly

Clinical Symptoms

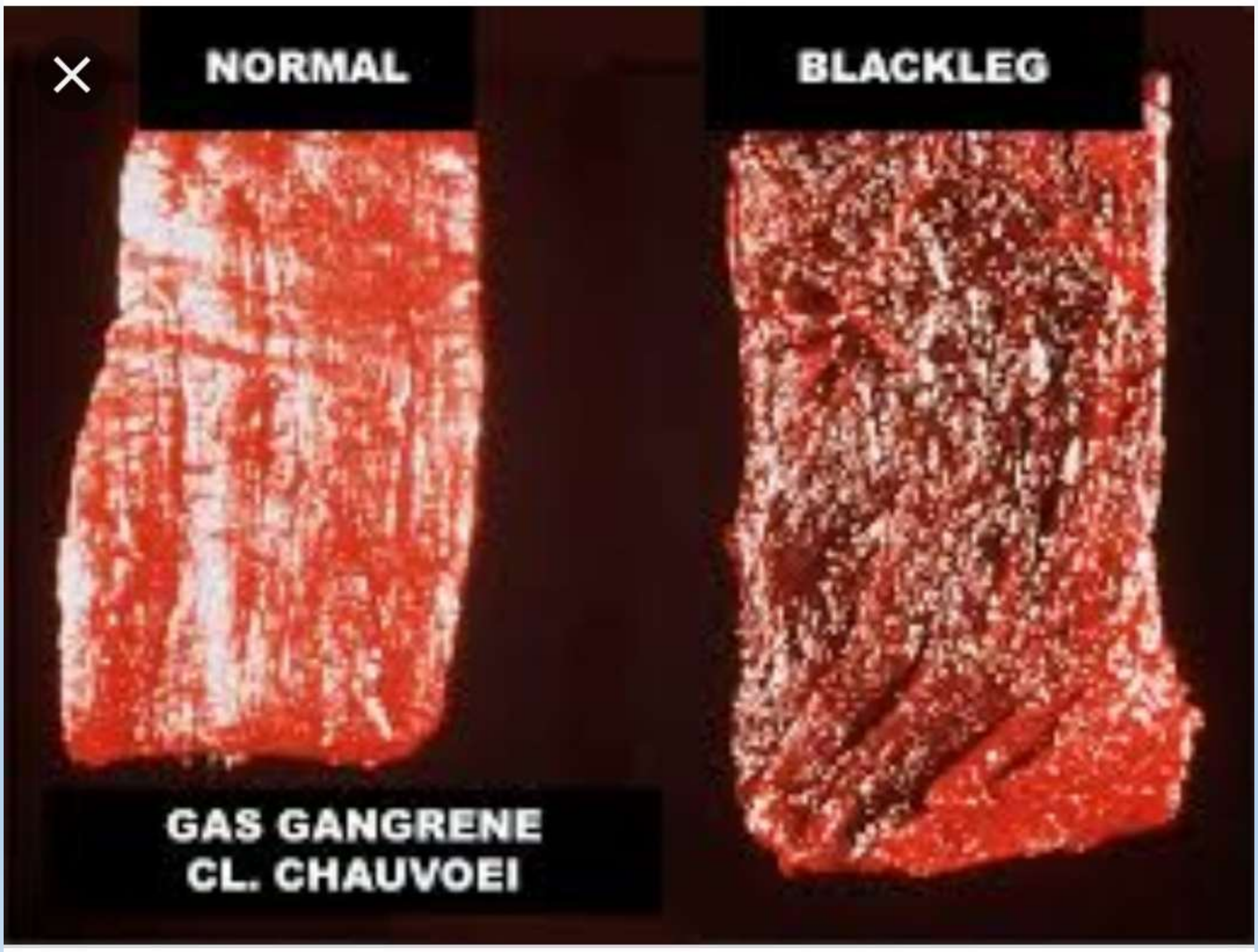
- Sometimes animal may be die without showing symptoms.
- The most obvious sign in a crepitate swelling in hind- or forequarters crackles when rubbed due to gas in the muscle.
- The symptoms are fever, lameness and switching of the muscles of the affected region.
- Death usually occurs within 24 hours of the symptoms first observed.
- The affected region is hot and painful but soon becomes cold and painless, and there is crepitation due to gas.
- The skin over the affected area becomes dry, hard and dark.
- Sometimes the muscle of neck and back is affected in sheep; there is high fever and anorexia.

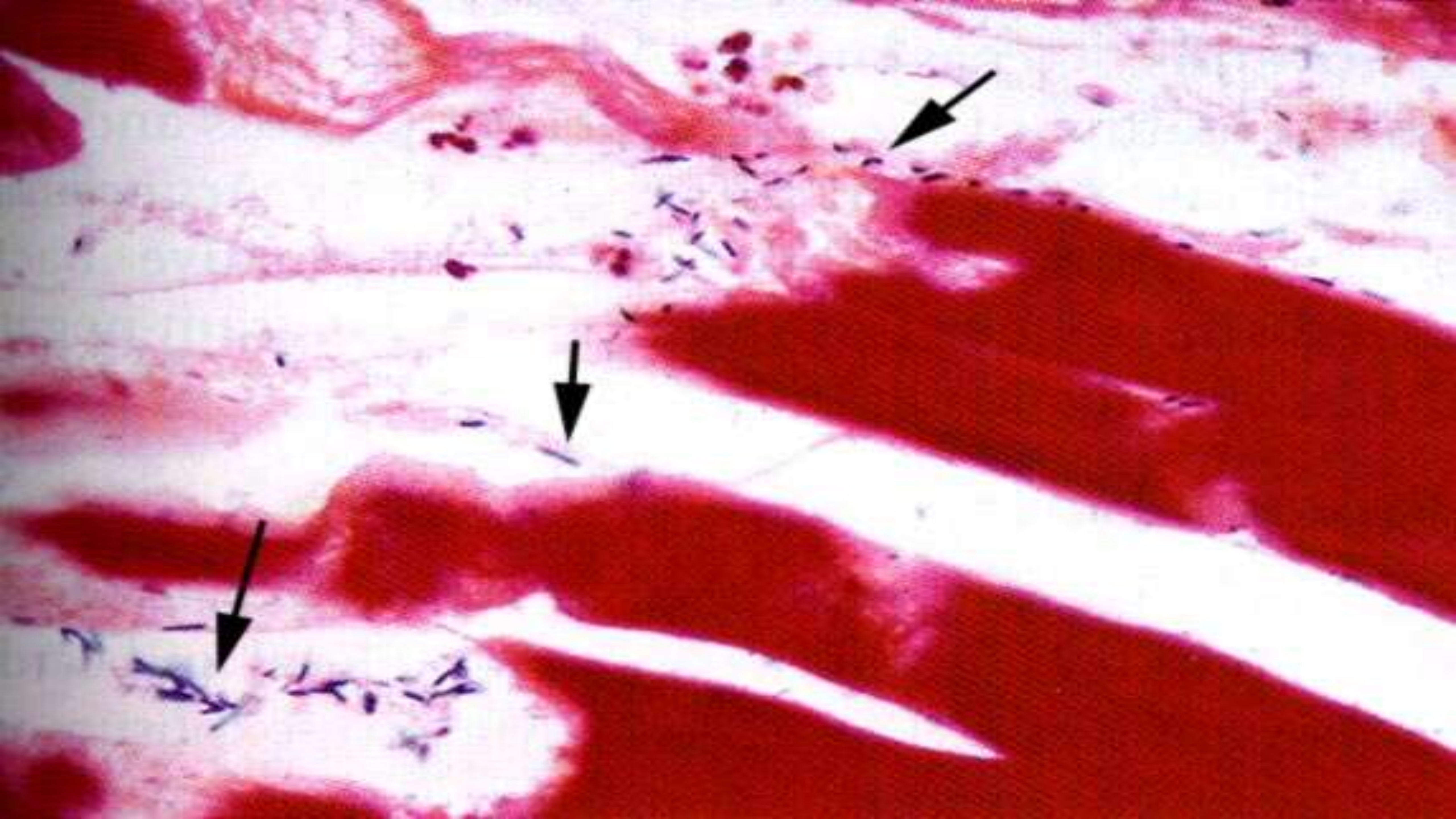
Macroscopic lesions

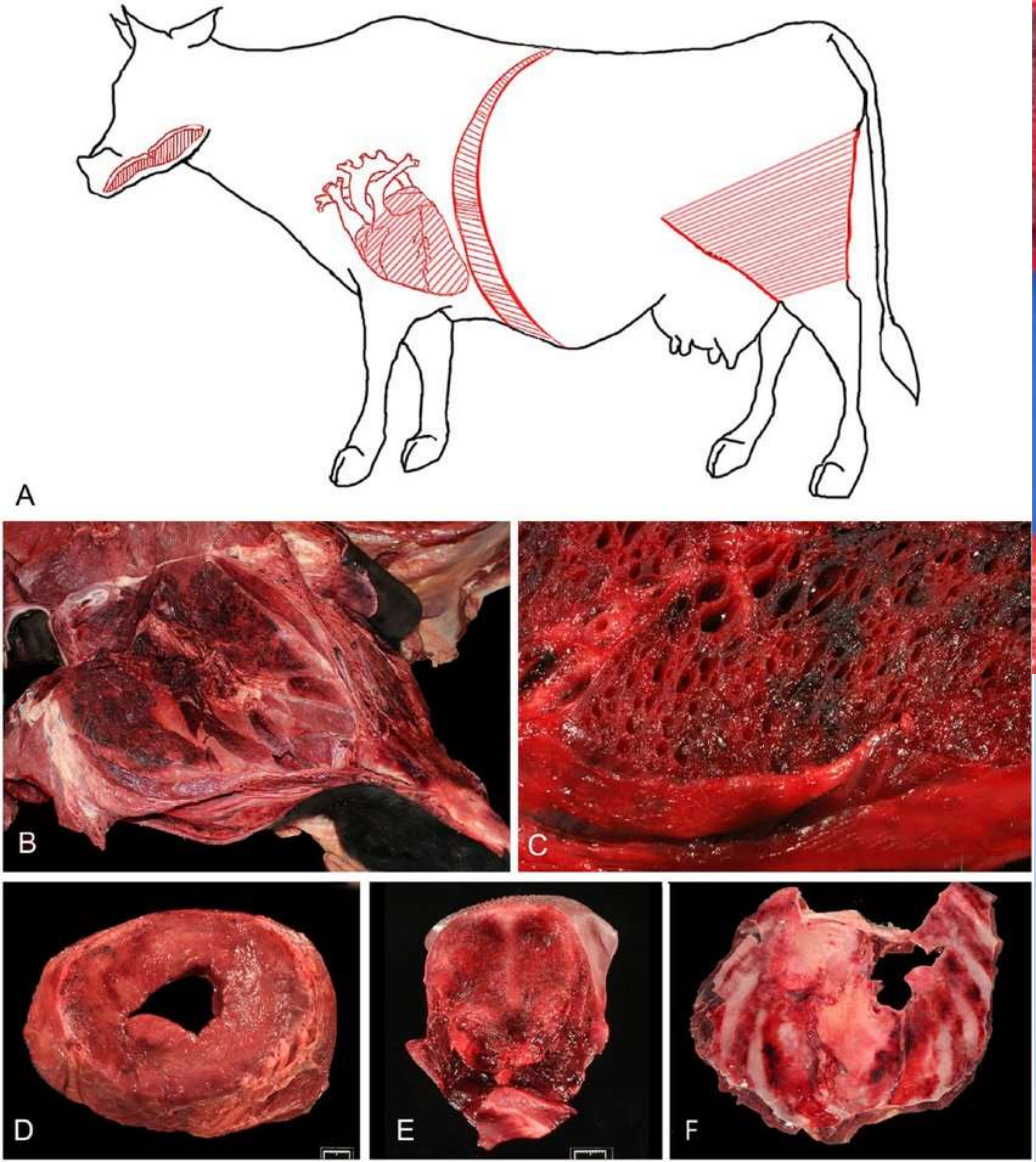
- Emphysematous necrotic myositis.
- In the central part of the lesion there is usually a well-defined area of muscle, which is dark red in color, dry, necrotic and filled with small gas bubbles, which give a swollen appearance to the muscles. The lesion has a characteristic rancid odor.
- Surrounded this area of muscle there will be yellowish or blood stained edematous fluid.
- In ewes there will be necrosis of the vaginal mucosa and skin with extensive edema involving the hind limbs and thigh muscles.

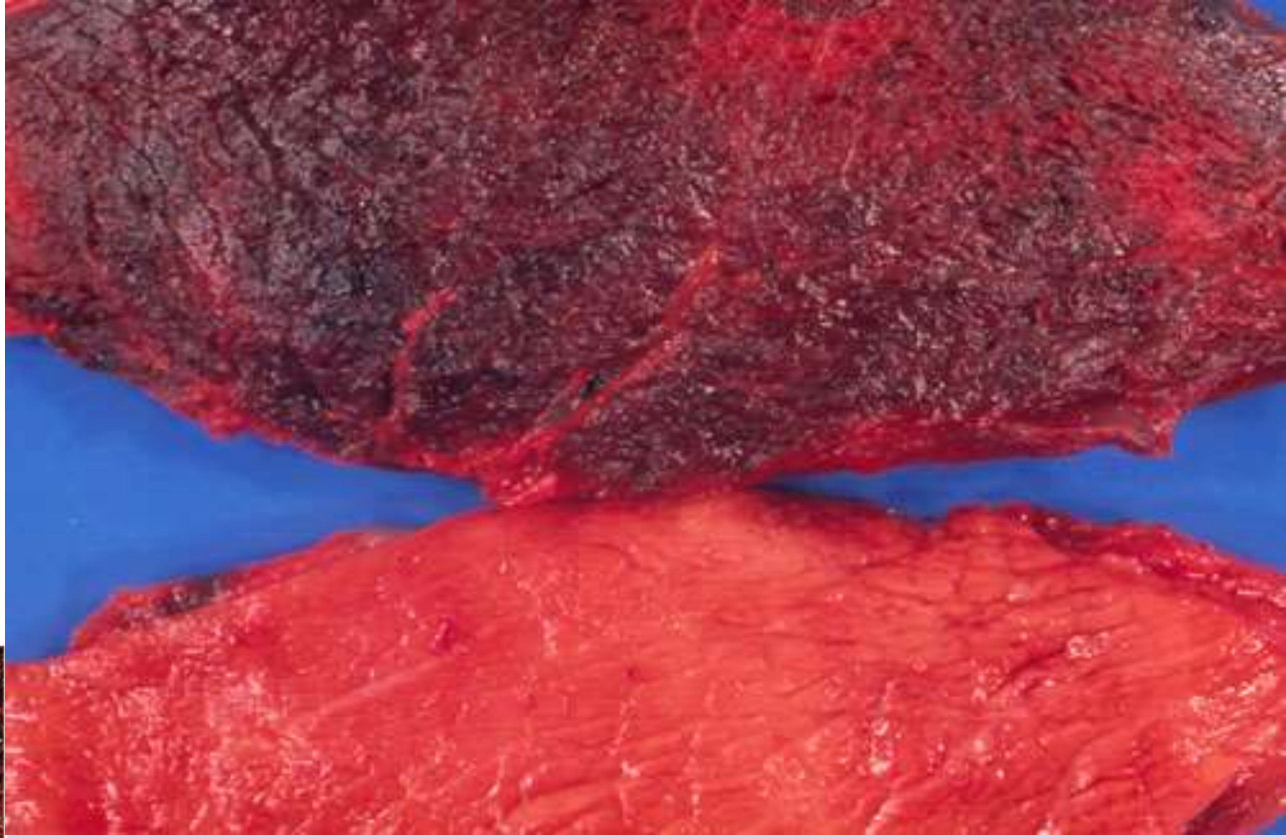
MACROSCOPIC LESIONS

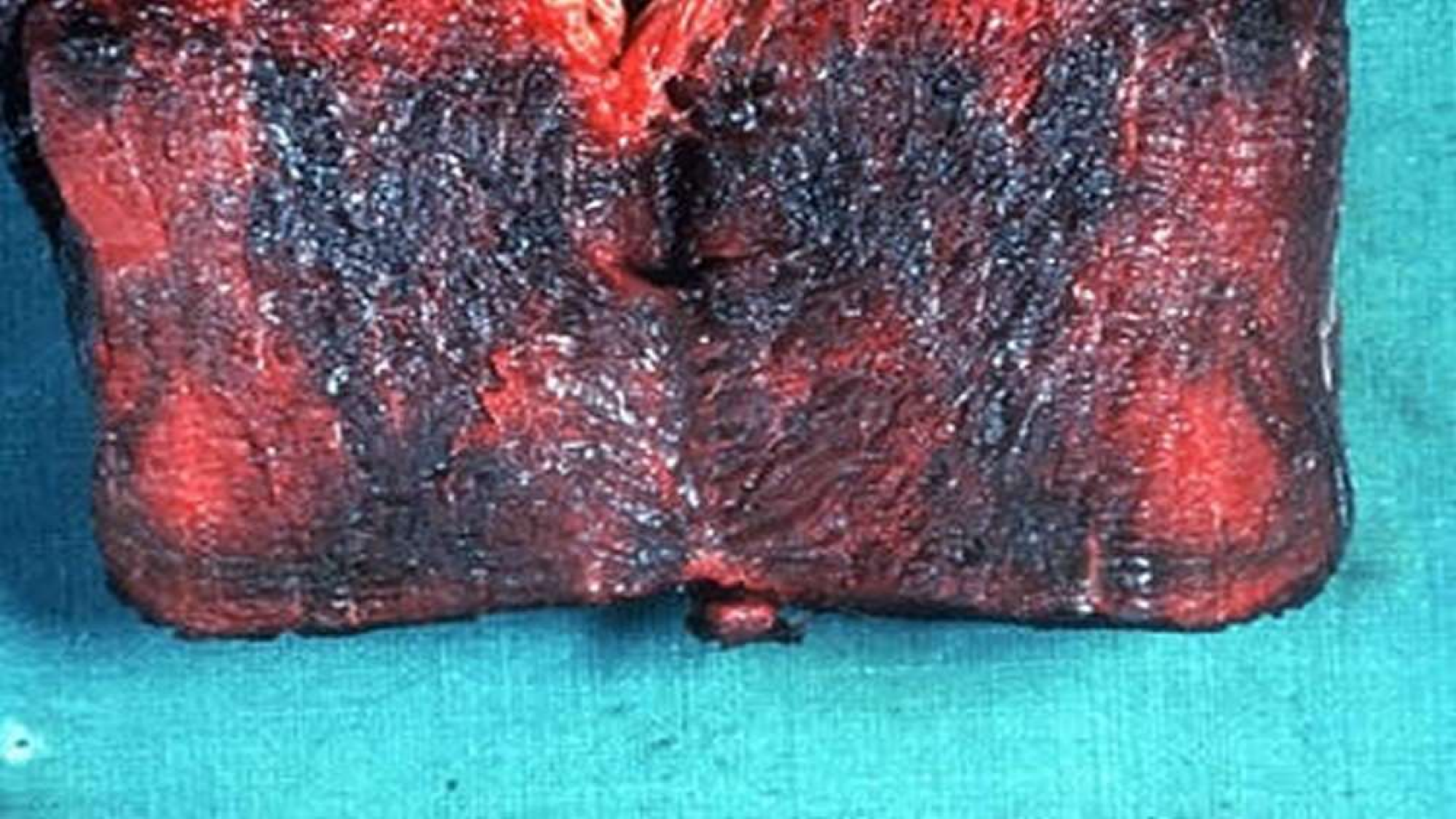
- Emphysematous Necrotizing Myositis.
- Fibrino-Hemorrhagic Pericarditis ,Pleurisy and Ulcerative Endocarditis.
- Blood stained edematous fluid.
- Rancid odor, swollen due to gas inside.


NORMAL
BLACKLEG
GAS GANGRENE
CL. CHAUVOEI

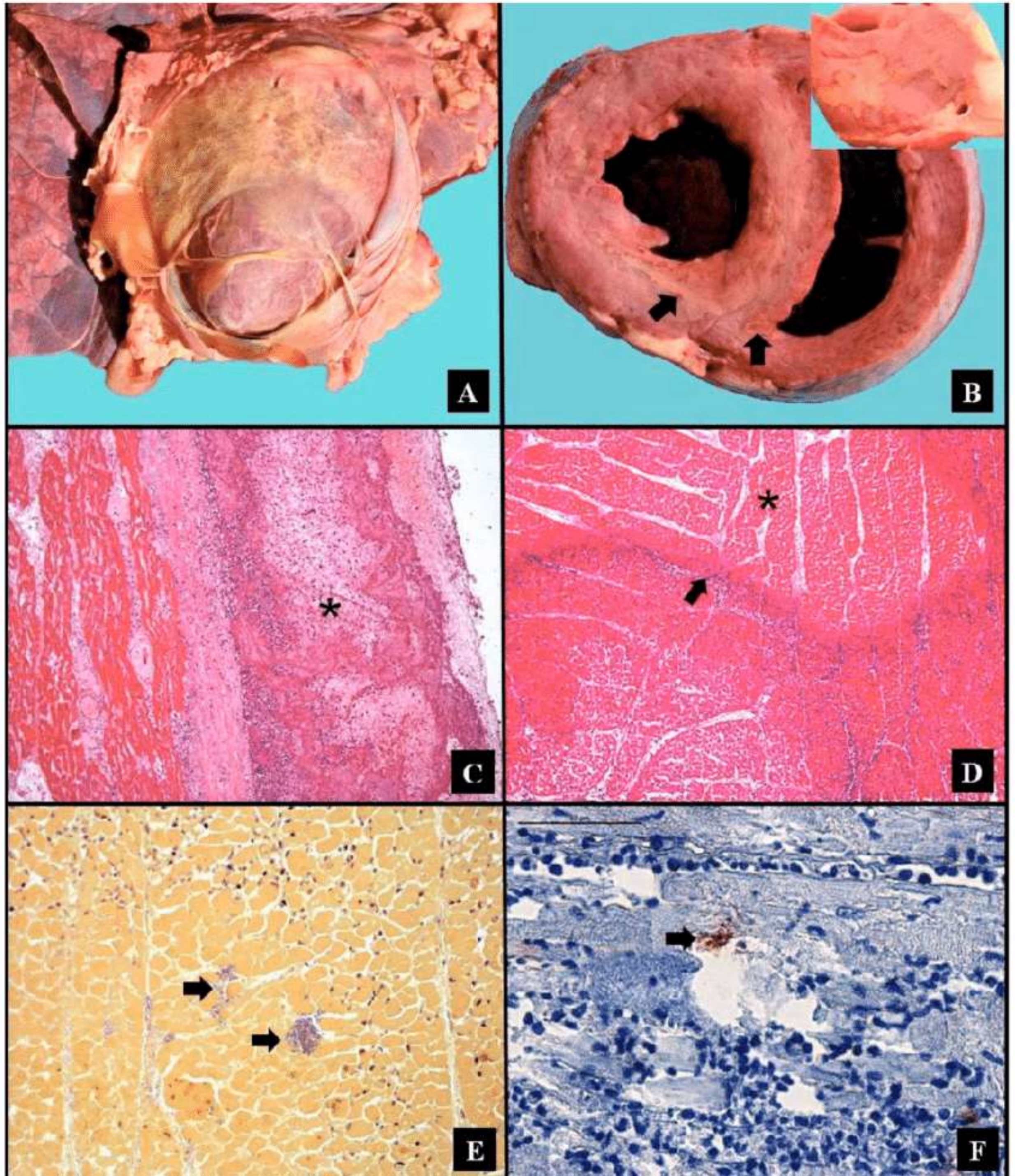



A
B
C
D
E
F





Clostridium chauvoei **in a Gram-stained smear and the scanning EM**



MYOCARDITIS COMMON IN BLACK QUARTER

MICROSCOPIC LESIONS

- Degeneration and coagulative necrosis of muscle
- Streaks of hemorrhage around muscle fibers
- Infiltration of WBC and
- Rod Shaped bacteria will be present.

DIAGNOSIS

- Based on history and symptoms - The most obvious sign is crepitate swelling particularly in the hind or fore quarter which rackels when rubbed with the fingers as a result of gas production.
- Materials to be collected include muscle, edema fluid and bone marrow from long bone.
- Smears prepared from the lesions and edematous fluid reveal Gram positive rod/ pleomorphic bacteria.
- Isolation can be done from the center of the lesion, edematous fluid and from heart blood & spleen.
- FAT used to differentiate from *C. septicum*

DIAGNOSIS

- Broth cultures or edematous fluid from the lesions can be tested for toxicity and specific neutralization by antitoxin in mice or guinea pigs.
- Lab animals of choice are hamsters and guinea pigs.
- The muscle tissue is collected, dried, triturated with normal saline, filtrate is prepared and heat it at 650C for 30 min to destroy vegetative bacteria (spores will remain).
- Inoculate 1 ml of filtrate with I ml of 10% CaCl2 i/m in guinea pig (CaCl2 is an irritant which cause injury to tissue so that spores can germinate).
- PCR

TREATMENT PROTOCOL

- Incise swelling and remove exudates.
- Early therapy with antibiotics is effective and beneficial.
- The affected animals should be immediately treated with Penicillin @ 6,000 IU/kg body weight by IM route for protection against the disease.
- Penicillin-G is the drug of choice.
- Antihistaminic injection like Pheniramine maleate or Chlorpheniramine maleate may be administered.
- Anti-inflammatory analgesic injection like Meloxicam.
- Ketoprofen may be administered as painkiller.

Prevention and Control

- Hyper immune serum (HIS) is used to control explosive outbreaks. Penicillin along with HIS is used to treat the disease.
- Oxytetracycline & Chlortetracycline can also be employed effectively in early stages.
- Vaccination- all the cattle in area are to be vaccinated at 3-6 month of age and then annually. Ewes are vaccinated 3 weeks prior to parturition
- The most reliable results are obtained from using a formalized alum precipitated whole culture that confers immunity against the bacteria as well as the toxin.
- For economic reasons a multi-component vaccine containing *C. chauvoei, C. septicum, C.welchii* type D and *C. tetani* is used.
- A stronger immunity is stimulated by two doses of vaccine at a time interval of at least 2-3 weeks
- Management practices like change in pasture can be tried.

VACCINATION

- Prevention is possible by routine vaccination once in a year.
- Combined HS BQ vaccine @ 2 ml of Intervet and 4 ml of Biomed company per animal by SC route can be given.
- Bovilis HS BQ (Intervet) 50 ml (25 dose vial) may be given to cattle and buffalo @ 2 ml SC injection


HAEMORRHAGIC SEPTICAEMIA
AND BLACK QUARTER VACCINE
Raksha-H.S.+B.Q.
हेमोरेजिक सेप्टिसेमीया और
ब्लैक क्वार्टर मिश्रित टीका
रक्षा-एच.एस+बी.क्यू
To be protected from light.
Store between 2°C and 8°C.
Avoid freezing.
Shake well before use.
2023D01S2

VACCINE

FACTS AND TIPS.

- Spores are oval, located sub-terminally and wider than bacilli (tennis racquet) appearance.
- Sometimes pleomorphic, large cigar shaped rods, pear shaped, navicular or citron (lemon) shaped forms occur.
- In cooked meat medium growth is slow and meat is turned pink (Saccharolytic). Organism produces cloudiness with accumulation of air bubbles and culture is having a rancid odor.
- Spores are destroyed by formalin and per chloride of mercury.

A case In South Africa

- In 2014,35 rhinoceros were found dead in a farm run by John Hume, on black quarter outbreak.
- Carcass of dead one were burned to ash to prevent the disease from spreading the remaining 65.
- 67% of the disease were acute form.

THANK YOU